Status	Finished
Started	Sunday, 11 October 2026, 11:23 PM
Completed	Sunday, 11 October 2026, 11:25 PM
Duration	2 mins
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are ignored
b. Each part is searched for a specific element
c. The parts are merged without sorting
d. Each part is sorted independently

The correct answer is: Each part is sorted independently

Question **2**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It has no effect on the search operation
b. It allows for faster searching
c. It increases the number of comparisons needed
d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Binary Search
b. Bubble Sort
c. Linear Search
d. Merge Sort

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Dynamic programming
b. Greedy method
c. Backtracking
d. Divide and conquer

The correct answer is:
Divide and conquer

Question **5**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It avoids the need for recursion
b. It allows for efficient parallel processing and sorting of data
c. It only works on small datasets
d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **6**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes code execution slower
b. Decreases the efficiency of algorithms
c. Makes data harder to manage
d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **7**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are discarded
b. They are split again into smaller sublists
c. They are combined to form a single sorted list
d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **8**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 80,150,200,470,40,90,300,400,70,120
b. 200,470,80,150,40,90,300,400,70,120
c. 40,80,90,150,200,300,400,470,70,120
d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **9**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Insertion
- b. Distribution
- c. Extraction
- d. Selection

The correct answer is:
Insertion

Question **10**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Complexity
- b. Merging
- c. Sorting
- d. Searching

The correct answer is:
Complexity

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Quick Sort
- b. Merge Sort
- c. Insertion Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **12**

Complete

Mark 0.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Bubble Sort
- b. Merge Sort
- c. Quick Sort
- d. Heap Sort

The correct answer is: Bubble Sort

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting decreases the efficiency of selection operations
- b. Sorting helps in finding duplicates quickly
- c. Sorting is rarely used in programming
- d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly

Question **14**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Merge Sort
- b. Binary Search
- c. Linear Search
- d. Bubble Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It slows down the process
c. It makes it easier to select items based on their relationship to the rest of the items
d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items